Quantification: 699_HYP_HP_z11c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	22
interior lacuna count	17
frame-edge lacuna count	5

Interior lacunae only: mean, median and sample SD (n = 17)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2623.41	1403	3027.55	px^2
major_axis_length_px	lacuna shape	87.099	67.1123	52.4465	px
minor_axis_length_px	lacuna shape	42.1822	34.896	20.5754	px
aspect_ratio	lacuna shape	2.1526	1.7046	0.9474	unitless
eccentricity	lacuna shape	0.7999	0.8098	0.1517	unitless
solidity	lacuna shape	0.7114	0.7052	0.0887	unitless
roots_count	headline	9.8235	11	5.1019	count
ring_length_r30_px	headline	381.588	360	206.266	px
ring_length_r60_px	reported	682.765	604	403.524	px
owned_length_px	ownership-dependent	842.93	727.813	751.557	px
edge_count	ownership-dependent	46.0588	35	43.1197	count
mean_edge_length_px	ownership-dependent	19.2843	19.2782	2.8298	px
perimeter_px	normalised	337.356	234.25	222.27	px
ring_area_r30_px2	normalised	9639.35	8057	4586.04	px^2
ring_area_r60_px2	normalised	21002.3	19153	9730.96	px^2
in_frame_fraction_r30	normalised	0.994	1	0.0205	unitless
in_frame_fraction_r60	normalised	0.9799	1	0.0568	unitless
ring_density_r30	normalised	0.040822	0.0425232	0.0154931	px^-1
ring_density_r60	normalised	0.0326193	0.032632	0.0116929	px^-1
roots_per_100px_perimeter	normalised	3.2788	3.4555	1.6568	per 100 px
ring_attached_length_r30_px	network v2	331.882	317	216.325	px
ring_attached_length_r60_px	network v2	517.647	473	392.356	px
ring_length_w_r30_px	network v2	429.627	413.99	231.119	px
ring_length_w_r60_px	network v2	765.107	689.713	448.788	px
sholl_crossings_r10	network v2	9.1176	8	5.6886	count
sholl_crossings_r20	network v2	9.2941	10	4.4267	count
sholl_crossings_r30	network v2	9.4118	9	4.7442	count
loop_count	loops, report only	2.4706	0	4.0018	count
loop_edge_share	loops, report only	0.1115	0	0.1385	share
ring_width_mean_r30_px	width	7.4834	7.6788	1.0406	px

Per field
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measure	value	units
analysed_area_px2	965943	px^2
total_skeleton_length_px	22219	px
canalicular_length_density_per_px	0.0230024	px^-1
junction_count	500	count
junction_density_per_px2	0.00051763	px^-2
lacuna_count	22	count

lacunae_per_px2	2.278e-05	px^-2
skeleton_component_count	318	count
mean_component_length_px	69.8711	px
median_component_length_px	15	px
field_length_density_w_per_px	0.0257026	px^-1
field_density_without_flagged_per_px	0.0222516	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	603	px
unattached_component_count	31	count
canalicular_length_density_attached_per_px	0.0230024	px^-1
field_length_density_with_unattached_per_px	0.0236266	px^-1
loop_count	62	count
loop_edge_share	0.1923	share
edge_count	1305	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.6245	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	17321	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	483	34.5143	18.5791	1.8577	0.8428	0.9028	-1.3729	9.6004	721.977	True
2	889	60.8664	21.5104	2.8296	0.9355	0.7566	-1.1858	41.9595	618.117	False
3	971	48.2689	39.9508	1.2082	0.5612	0.6637	-1.1736	126.652	613.53	False
4	1293	67.1123	29.3143	2.2894	0.8996	0.7085	-0.9693	147.124	738.418	False
5	933	52.1712	30.6068	1.7046	0.8098	0.7272	-0.9318	146.222	325.69	False
6	8385	126.932	101.718	1.2479	0.5982	0.7052	-1.4611	277.754	204.132	False
7	1905	74.7758	58.1104	1.2868	0.6293	0.5641	1.396	301.221	621.288	False
8	9083	140.536	109.35	1.2852	0.6281	0.6544	-0.1917	390.607	39.1286	True
9	11096	170.603	105.697	1.6141	0.785	0.7494	-0.1521	533.955	42.6183	True
10	1403	54.0186	34.896	1.548	0.7633	0.888	-0.1596	368.566	862.24	False
11	3093	92.0714	54.1062	1.7017	0.8091	0.6916	1.2856	410.259	393.304	False
12	1944	82.0911	39.6724	2.0692	0.8755	0.6506	1.2002	562.526	659.101	False
13	1195	44.9336	38.8022	1.158	0.5043	0.7607	-1.4455	562.059	755.791	False
14	2018	112.423	30.231	3.7188	0.9632	0.6731	1.4432	571.53	456.488	False
15	634	48.158	30.1352	1.5981	0.78	0.5992	-1.5103	631.199	551.279	False
16	1329	62.4213	28.6247	2.1807	0.8887	0.8819	1.2484	643.479	606.14	False
17	812	43.7603	30.951	1.4139	0.7069	0.7167	-1.4386	655.233	193.442	False
18	12055	251.462	78.5981	3.1993	0.9499	0.6372	-0.5972	767.904	745.585	False
19	2798	118.207	35.7656	3.305	0.9531	0.8127	-1.3169	814.447	954.982	False
20	2396	80.3638	41.3693	1.9426	0.8573	0.8812	-1.0185	852.972	994.821	True
21	14977	374.03	83.4021	4.4847	0.9748	0.5035	-1.4304	910.923	184.857	True
22	2941	141.011	34.1041	4.1347	0.9703	0.6575	1.2599	895.57	423.196	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	1	98	161	41.1421	1	41.1421	5.5126
2	4	203	249	171.569	11	15.5971	7.3657
3	10	358	604	725.027	44	16.4779	8.6418
4	12	386	697	821.826	35	23.4807	8.1517
5	8	302	589	854.311	54	15.8206	9.3181
6	18	850	1507	1708.14	104	16.4244	7.7688
7	14	555	1208	1423.49	89	15.9943	6.9678
8	17	546	744	648.772	50	12.9754	8.1185
9	10	245	370	335.978	18	18.6654	8.7693
10	2	209	470	315.622	15	21.0415	6.8106
11	18	813	1542	3050.08	170	17.9416	7.4887
12	11	392	529	520.99	32	16.2809	7.8859
13	3	166	268	87.0122	4	21.753	5.0802
14	13	424	731	727.813	35	20.7947	7.4846
15	11	235	335	925.352	48	19.2782	6.3267
16	4	325	421	208.238	11	18.9307	7.8808
17	10	360	625	789.713	37	21.3436	8.5944
18	12	411	688	312.078	13	24.006	5.9023
19	3	66	220	153.468	7	21.924	7.6788
20	0	0	0	0	0	0	
21	20	603	1219	2081.06	117	17.7868	8.5339
22	14	432	924	1535.08	74	20.7444	7.8714

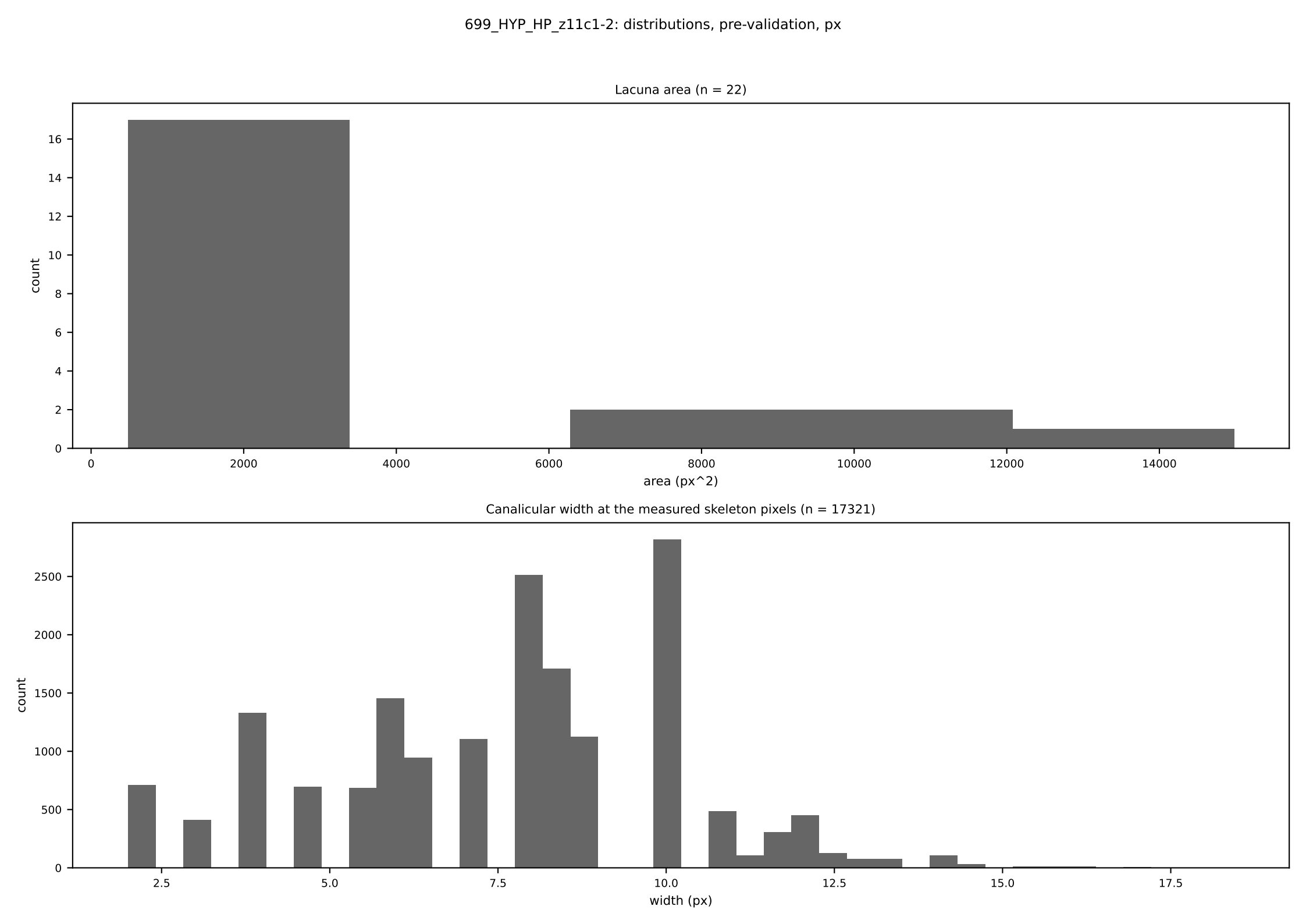
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	95.2548	3348	7313	0.6198	0.5747	0.0292712	0.0220156	1.0498
2	172.995	6815	11086	0.9811	0.8306	0.0297872	0.0224608	2.3122
3	191.622	7249	15098	1	1	0.0493861	0.0400053	5.2186
4	234.25	8068	19697	1	1	0.0478433	0.0353861	5.1227
5	172.995	7102	18424	1	1	0.0425232	0.0319692	4.6244
6	894.833	17009	33459	1	1	0.0499735	0.0450402	2.0115
7	394.333	10570	25661	1	1	0.0525071	0.0470753	3.5503
8	756.861	10512	16869	0.6862	0.6501	0.0519406	0.0441046	2.2461
9	695.896	10926	20640	0.7085	0.658	0.0224236	0.0179264	1.437
10	174.066	7351	20270	1	1	0.0284315	0.023187	1.149
11	467.446	11796	28229	1	1	0.0689217	0.0546247	3.8507
12	318.333	9236	16615	1	1	0.0424426	0.0318387	3.4555
13	232.285	6906	15617	1	1	0.0240371	0.0171608	1.2915
14	409.931	10384	22886	1	1	0.0408321	0.0319409	3.1713
15	174.853	5222	9580	1	1	0.0450019	0.0349687	6.291
16	172.267	5412	9650	1	1	0.0600517	0.0436269	2.322
17	182.894	6958	19153	1	1	0.051739	0.032632	5.4676
18	834.833	23261	48925	1	1	0.0176691	0.0140623	1.4374
19	304.777	8057	16374	0.9164	0.8272	0.00819163	0.0134359	0.9843
20	235.48	4021	7954	0.6927	0.6449	0	0	0
21	1353.43	28693	52942	0.9016	0.8594	0.0210156	0.0230252	1.4777
22	402.333	12473	26315	1	1	0.0346348	0.035113	3.4797

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	85	99	105.184	173.64	4	2	2
2	99	103	225.924	274.38	5	6	8

3	329	496	413.99	689.713	8	8	10
4	374	621	434.049	768.115	9	12	12
5	272	545	349.907	668.215	6	9	7
6	826	1265	951.21	1682.87	25	16	15
7	526	994	621.002	1339	14	12	17
8	513	595	620.215	850.352	12	14	8
9	233	277	278.108	422.061	7	7	2
10	169	284	233.752	529.76	4	4	7
11	799	1489	916.37	1721.76	18	18	17
12	333	413	439.948	597.002	10	10	9
13	92	92	186.024	295.38	5	4	4
14	400	491	468.605	820.754	11	13	7
15	207	224	264.309	374.978	5	5	4
16	317	364	377.019	482.446	8	8	4
17	321	490	406.563	701.6	8	11	9
18	246	334	462.688	782.637	8	11	10
19	61	122	72.6985	249.066	1	1	3
20	0	0	0	0	0	0	0
21	438	1131	656.103	1351.76	15	20	15
22	271	473	479.605	1029.15	10	10	17


699_HYP_HP_z11c1-2: distributions, pre-validation, px
Lacuna area (n = 22)
count
area (px^2)
0
2
4
6
8
10
12
14
16
0
2000
4000
6000
8000
10000
12000
14000
Canalicular width at the measured skeleton pixels (n = 17321)
count
width (px)
0
500
1000
1500
2000
2500
2.5
5.0
7.5
10.0
12.5
15.0
17.5